PROJECT REPORT

FORMULA 1 RACE INTELLIGENCE ANALYSIS

A Tableau-based visual analytics project for exploring Formula 1 race, driver, team and circuit performance.

Project detail	Information
Project name	Formula 1 Race Intelligence Analysis
Technology indicated by project materials	Tableau; Snowflake performance-test template supplied
Project date	To be confirmed
Team ID	PNT2022TMIDxxxxxx (replace with confirmed team ID)
Prepared by	Project team (add member names)

Prepared using the supplied project overview, circuit dataset extract and report format. Fields requiring evidence not present in the supplied materials are clearly marked for completion.

1. INTRODUCTION

1.1 Project Overview

Formula 1 Race Intelligence Analysis is a visual analytics project intended to help users explore patterns and trends in the sport. The supplied overview describes analysis of driver performance, team championship trajectories, tyre-compound strategies, circuit-level race participation, driver reliability and Did Not Finish (DNF) patterns, and global podium achievements. Tableau is identified as the tool used to present interactive visualizations, dashboards and a story.

The overview describes a broader analytical dataset of 6,464 race records across 17 seasons. The separately supplied circuit extract contains 29 circuit records and 9 fields: circuit_id, circuit_name, country, city, laps, lap_km, type, altitude_m and direction. These are different levels of data and should not be treated as the same record count.

1.2 Purpose

The purpose is to convert Formula 1 data into accessible visual insights that support comparison, exploration and evidence-based discussion. The project is designed for three user scenarios: a team performance strategist benchmarking performance and tyre strategies; a broadcast analyst exploring driver rankings, host countries and podium distributions; and a technical regulation analyst investigating DNF patterns and driver reliability.

2. IDEATION PHASE

2.1 Problem Statement

Formula 1 performance information spans seasons, drivers, teams, circuits and race outcomes. Reviewing these dimensions separately can make it difficult for strategists, journalists and technical analysts to identify trends or compare performance consistently. The project addresses this problem by organizing relevant data into interactive Tableau views so users can filter, compare and interpret key performance indicators.

2.2 Empathy Map Canvas

User group	Thinks / needs	Sees / does	Pain points
Team performance strategist	Needs to benchmark team points, evaluate tyre choices and identify talent.	Compares championship trends, circuits, compounds and driver performance.	Information is spread across multiple dimensions; comparisons may take time.
Broadcast analyst	Needs fast, clear statistics and visuals for pre-race and post-race coverage.	Looks up driver points, host-country counts and podium distributions.	Needs concise, trustworthy visuals that can be understood quickly.
Technical regulation analyst	Needs to understand DNF causes and reliability patterns.	Studies failure categories, drivers, teams, circuits and season trends.	Failure patterns may be difficult to detect across many races and seasons.

2.3 Brainstorming

The project overview suggests the following dashboard ideas: (1) championship-points trajectory comparison; (2) tyre-compound usage and outcome comparison by circuit type; (3) driver performance leaderboard; (4) Grand Prix host-country frequency; (5) podium distribution by driver and team; (6) DNF and failure-category analysis; and (7) driver reliability comparison across seasons. The circuit extract can additionally support circuit profile comparisons by country, type, lap count, lap length, altitude and direction.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User action	Dashboard support	Expected outcome
Identify question	Select a scenario and KPI.	Scenario-oriented dashboard/story navigation.	Clear analytical objective.
Explore	Choose season, driver, team, circuit or compound.	Interactive filters and chart selections, where implemented.	Focused subset of data.
Compare	Compare values and trends across categories.	Rankings, trend charts, distributions and comparison views.	Visible differences and patterns.
Interpret	Review chart labels, tooltips and context.	Titles, legends, labels and explanatory text.	Meaningful interpretation.
Communicate	Use a view in a report or broadcast.	Dashboard/story presentation and export capability, if enabled.	Shareable evidence and insights.

3.2 Solution Requirement

Functional requirements: load the relevant F1 tables; relate race, driver, constructor/team, circuit and result information using valid keys; provide views for the three scenarios; allow users to filter the dimensions relevant to each view; calculate required measures such as points, podium counts and DNF counts; and present results through dashboards and a Tableau Story.

Non-functional requirements: accurate and consistent data; readable labels and legends; responsive filter interactions; appropriate aggregation; clear units; and documented limitations. Actual response-time targets and measured performance values were not included in the supplied materials.

3.3 Data Flow Diagram

Step	Data flow
1. Source	F1 race dataset and circuit reference data are collected.
2. Preparation	Fields are checked, cleaned, standardized and assigned appropriate data types.
3. Storage / connection	Prepared data is made available to the analytics layer. Snowflake is referenced by the supplied Snowflake performance-test template; the actual connection configuration was not supplied.
4. Analysis	Tableau connects to the prepared data, relates tables and creates calculated measures.
5. Presentation	Interactive worksheets, dashboards and story points present the analysis to users.

3.4 Technology Stack

Layer	Technology / component	Role
Data source	F1 race dataset and circuit dataset extract	Provides race and circuit attributes for analysis.
Data warehouse	Snowflake (as indicated by the supplied Snowflake performance-test template)	Potential centralized storage/query layer; confirm actual deployment and connection.
Visualization	Tableau	Worksheets, calculated fields, dashboards, filters and story.
Documentation	Project report and performance-test templates	Records methodology, testing and outputs.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution aligns with the three stated user scenarios: team strategists need comparisons of championship performance and tyre strategies; broadcast analysts need quick leaderboards and podium/host-country summaries; and technical analysts need DNF and reliability views. A shared data model and scenario-specific dashboards can reduce repeated manual analysis and make comparisons easier to communicate.

4.2 Proposed Solution

Develop a Tableau workbook containing: (a) a team performance dashboard with championship points and tyre/circuit comparisons; (b) a broadcast analysis dashboard with driver leaderboard, host-country frequency and podium distributions; and (c) a technical analysis dashboard with DNF categories and reliability trends. Add filters relevant to each view and organize the key views into a Tableau Story. The circuit extract can support a circuit overview page.

4.3 Solution Architecture

Architecture layer	Description
Data inputs	Race, driver, constructor/team, result, tyre and circuit data. The supplied circuit extract has 29 rows and 9 columns.
Data preparation	Validate keys, types, nulls, duplicates and category consistency; document any exclusions.
Data platform	Snowflake layer if used in the implemented workflow; verify database, schema, tables and connection.
Analytics layer	Tableau data model, relationships/joins, calculated fields and worksheets.
Presentation layer	Scenario dashboards, filters, tooltips and Tableau Story.
Users	Team strategist, broadcast analyst and technical regulation analyst.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Planned activities	Deliverable
1. Scope and ideation	Confirm users, questions, KPIs and project boundaries.	Problem statement and scenario definitions.
2. Data understanding	Inspect tables, fields, keys, date ranges and record counts.	Data inventory and data dictionary.
3. Data preparation	Clean fields, standardize categories, validate nulls and duplicates.	Prepared and documented dataset.
4. Data modelling	Define relationships/joins and measures.	Analysis-ready data model.
5. Visualization	Build worksheets, filters, dashboards and story points.	Tableau workbook.
6. Testing	Check correctness, filters, rendering and query performance.	Completed test evidence.
7. Finalization	Capture screenshots, write report and prepare demo.	Final report and presentation.

A dated schedule, task owners and actual completion status were not supplied. Add the team's approved dates and member responsibilities before submission.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The supplied performance-test templates identify the testing areas below. The project overview confirms Tableau visualizations, dashboards and a Story were used. The actual workbook, Snowflake query history and measured

execution timings were not supplied; therefore, no response-time or query-performance result is claimed in this report.

Test area	What to record / verify	Current evidence status
Data rendered	Tables loaded, record counts and data size. Circuit extract: 29 records, 9 columns.	Circuit extract count verified from supplied PDF; full race-table counts need workbook/source confirmation.
Data preprocessing & quality	Null checks, duplicate checks, trimming, standardization, type checks and cleaning log.	Steps to be completed and evidenced with actual results.
New / calculated fields	List each calculated field, formula, data type and purpose.	Actual Tableau calculated-field list not supplied.
Filter efficiency	List filters and compare interaction/query performance with filters on/off or different selections.	Requires measured workbook/Snowflake test evidence.
Dashboard & visualization testing	Count worksheets/graphs and test rendering, labels, tooltips and interactions.	Actual visualization count and screenshots not supplied.
Query history	Identify the query with the highest execution time and record duration.	Requires Snowflake query-history export or screenshot.

Suggested functional checks: confirm totals against source data; check that filters update all intended charts; verify that selecting one driver/team/circuit does not unintentionally remove relevant records; check null and unknown categories; and verify that dashboard titles, units and legends are understandable.

7. RESULTS

7.1 Output Screenshots

Insert genuine screenshots from the completed Tableau workbook in this section. The supplied image is a project-overview page, not an output dashboard, so it should not be presented as evidence of completed visualizations.

Screenshot slot	Required evidence to insert
Figure 1	Team Performance dashboard – championship points / team comparison.
Figure 2	Tyre Strategy analysis – compound usage or outcome by circuit.
Figure 3	Driver Performance Leaderboard – cumulative points by driver.
Figure 4	Podium / Grand Prix host-country analysis.
Figure 5	Driver Reliability / DNF analysis.
Figure 6	Tableau Story overview showing story points.

Results should be written only after reviewing the actual dashboard values. Include the main finding, the filters/seasons used and the corresponding screenshot for each result.

8. ADVANTAGES & DISADVANTAGES

Advantages	Disadvantages / limitations
Combines several F1 analysis scenarios in one reporting experience.	Accuracy depends on source completeness, consistent keys and correct data preparation.

Advantages	Disadvantages / limitations
Interactive views can help users explore drivers, teams, circuits and seasons.	Large or complex data models may affect rendering and filter performance.
Visual comparisons can make trends and outliers easier to communicate.	The supplied circuit extract alone cannot support driver, team, tyre or DNF analysis.
Scenario-based design serves strategic, broadcast and technical audiences.	Some performance claims require measured evidence from Tableau and Snowflake.

9. CONCLUSION

The Formula 1 Race Intelligence Analysis project proposes an interactive Tableau-based approach to exploring race, driver, team, tyre, circuit and reliability information. Its three scenarios provide a clear direction for team strategy, broadcast analysis and technical regulation review. The supplied circuit extract contributes circuit-level context, while the project overview describes a broader analysis covering 6,464 race records across 17 seasons. Final conclusions about championship trends, tyre effectiveness, podium leaders or DNF causes must be based on the completed workbook and validated source data.

10. FUTURE SCOPE

Future enhancements may include refreshing the dataset for additional seasons; adding more detailed pit-stop, lap-time, weather and telemetry data; incorporating predictive models for race outcomes or reliability; adding automated data refresh; improving mobile-friendly dashboard layouts; and publishing the workbook through Tableau Server/Cloud with role-based access. These are proposed extensions, not features confirmed in the supplied project materials.

11. APPENDIX

Source Code (if any)

No source-code files were supplied. If SQL scripts, Tableau calculated-field definitions, Python/R scripts or ETL code were used, attach them here with brief descriptions.

Dataset Link

Dataset supplied as: “F1 Racing Project Dataset.xlsx - Google Sheets.pdf”. The circuit extract contains 29 circuit rows and 9 fields. Add the original shareable dataset URL here: [Insert dataset link].

GitHub & Project Demo Link

GitHub repository: [Insert repository URL, or write “Not available”].
Project demo: [Insert Tableau Public / Drive / video URL, or write “Not available”].

Submission checklist

Item	Action before final submission
Team details	Replace placeholder team ID and add project member names.
Data evidence	Attach the full source table inventory, row counts and data-size evidence.
Testing evidence	Add actual data-quality results, calculated fields, filter timings and Snowflake query history.
Results	Replace screenshot slots with genuine dashboard and Story screenshots; add concise findings.
Links	Insert working dataset, GitHub and project demo URLs, if available.

Report prepared from the supplied project overview, circuit dataset extract and Final Report Template. Unsupported implementation details and measurements have been left for confirmation rather than fabricated.